Shreya Ann Sebastian

Data Science, Machine Learning and AI | MSc Computer Science, TU Delft

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Dutch citizen, no sponsorship needed | Available immediately | English (native), Dutch (professional working proficiency)

Profile

MSc Computer Science graduate (TU Delft, Artificial Intelligence track) with a BSc in Computer Science (Leiden, bioinformatics track), working on data science, machine learning and generative AI. Thesis work in unsupervised learning and statistical modelling on multimodal sensor data, and in 3D deep learning for image classification. Built and runs Tech Jobs Radar, a public job board at techjobsradar.nl, end to end from design to cloud deployment. Other work covers agentic AI (tool-calling LLM agents), customer segmentation and propensity modelling, computer vision and natural language processing. Experienced in explaining technical results to non-technical stakeholders.

Education

MSc Computer Science 2023 - 2026
Delft University of Technology Artificial Intelligence Technology Track
Thesis: Automated Discovery of Latent Activity Contexts from Multimodal Sensing for Peer-Interaction Analysis in Preschoolers with Hearing Loss.
Unsupervised pipeline on multimodal sensor data (UWB positioning and LENA audio): a Gaussian Mixture Model found activity contexts, Dominant Sets clustering detected spatial groups, and per-context linear mixed models revealed inclusion effects absent from the pooled analysis. Redefined the research question when the planned data became unavailable and presented results to developmental psychologists.
Supervision: Socially Perceptive Computing Lab, TU Delft, with external supervision from the University of Miami.

BSc Computer Science 2020 - 2023
Leiden University Bioinformatics Track
Thesis: Enhancing Pollen Classification Accuracy through the Use of Whole Stack Images in 3D Convolutional Neural Networks.
3D CNNs (custom ConvNet3D and ResNet3D) in PyTorch on microscope Z-stacks, with data augmentation, hyperparameter tuning and a whole-stack versus sub-stack comparison. Trained on only about 10% of the data used in earlier 2D-projection work, a ResNet3D trained from scratch reached 96.7% test accuracy, 2.7 percentage points under that work.

International Baccalaureate Diploma
International School Eindhoven
HL: Biology, Chemistry, Dutch B | SL: Environmental Systems & Societies, English A (Language & Literature), Mathematics

Projects

Tech Jobs Radar techjobsradar.nl
Personal project, deployed
Live job board that classifies tech roles and extracts structured fields from Dutch career sites. Designed, built and operated end to end on Kubernetes on AWS, with monitoring, alerting and automated evaluation tests in CI.
Tools: Python, FastAPI, Postgres, Kubernetes, AWS

RetroSTC-Net: Retroactive Consolidation for Neural Networks Repo
Personal research project, open source (MIT)
Retroactive weight consolidation inspired by synaptic tagging and capture, evaluated on Split-MNIST against EWC and SI.
Tools: Python, PyTorch

LLM-Panel Annotation Framework Repo

Group research project, TU Delft

Three-LLM panel that annotates meeting transcripts against a codebook and measures inter-model agreement.

Tools: Python, OpenAI, Gemini and Groq APIs

Online Retail Purchase Prediction Repo

Personal project

Databricks pipeline on real retail transactions with k-means customer segmentation and propensity models, tested on four later months with predictions set in advance. Boosted trees raised a category's average precision from 0.50 to 0.69.

Tools: PySpark, Databricks (Delta Lake, Unity Catalog, MLflow, Jobs), scikit-learn

Deep Learning Reproduction Project Repo

Group project, Deep Learning course, TU Delft

Reproduced a multi-view underwater image enhancement paper and wrote the blog post on what replicated.

Tools: Python, PyTorch

Fault-Tolerant Sensor Data Ingestion (Smart Home PoC) Repo

Software Architecture course, TU Delft

Sensor data ingestion validated under simulated network failures and high load, with CI that tests it in Docker.

Tools: Python, RabbitMQ, Pydantic, Docker, GitLab CI

Personal Finance Engine Repo

Personal project

Finds subscriptions in bank statements; a local tool-calling LLM agent names hidden merchants with cited evidence.

Tools: Python, DuckDB, RapidFuzz, Ollama

Skills

Programming: Python, R, MATLAB, SQL, JavaScript, Java, C++, object-oriented programming, algorithms and data structures

Data Science: Data analysis, probability and statistics, hypothesis testing, regression and linear mixed-effects models, feature engineering, clustering, customer segmentation, propensity modelling, experiment design, error analysis, predictive modelling, data mining, data visualisation; pandas, NumPy, scikit-learn, matplotlib

Machine Learning: Supervised and unsupervised learning, classification, deep learning (CNNs), computer vision, natural language processing, reinforcement learning, pattern recognition, Gaussian mixture models, continual learning, model evaluation (precision, recall, F1), ablation studies, explainability, probability calibration; PyTorch, TensorFlow, OpenCV, GPU training with CUDA

Generative AI: LLMs, agentic AI (tool-calling agents), prompt engineering, few-shot prompting, structured JSON outputs, information extraction, LLM evaluation, inter-annotator agreement; OpenAI, Gemini and Groq APIs, Ollama

Data Engineering: ETL and data pipelines, PySpark, Databricks (Delta Lake, Unity Catalog, Jobs), SQLAlchemy, PostgreSQL, SQLite, DuckDB, Redis queues, RabbitMQ, web crawling, deduplication, fuzzy text matching

Software & MLOps: REST APIs (FastAPI), Pydantic, MLflow (experiment tracking, model registry), Docker, Kubernetes (k3s, Helm), Terraform, AWS (EC2, S3, IAM), CI pipelines (GitHub Actions, GitLab CI), evaluation and regression testing in CI, unit testing (pytest), monitoring and alerting (Prometheus), Cloudflare, Git

Domains: Human-computer interaction, bioinformatics

Communication & Collaboration: Presenting technical results to non-technical audiences, stakeholder alignment, creative problem-solving, technical writing, team projects, Agile, Scrum

Extracurriculars: Basketball at DSBV Punch (TU Delft) and varsity level (International School Eindhoven), Student Council, Science Club